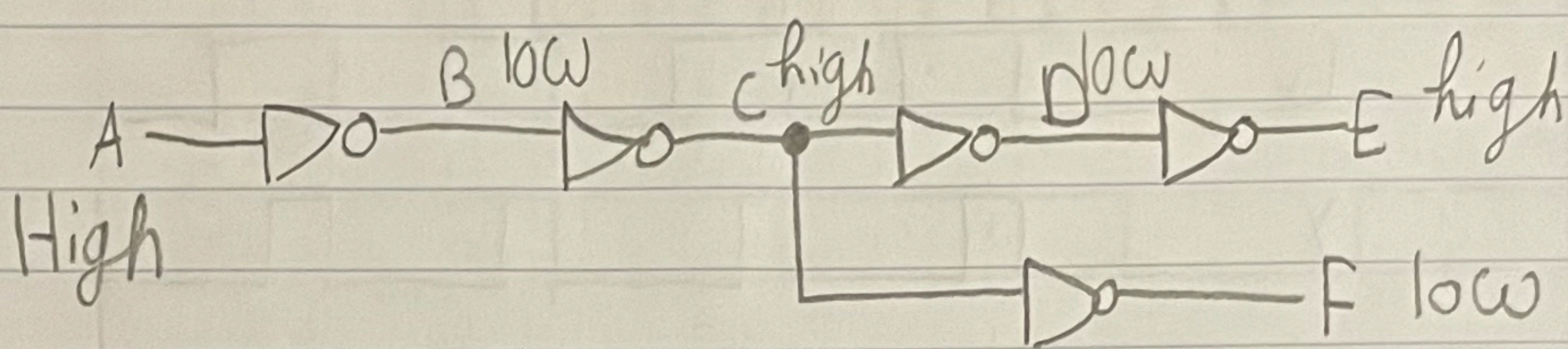


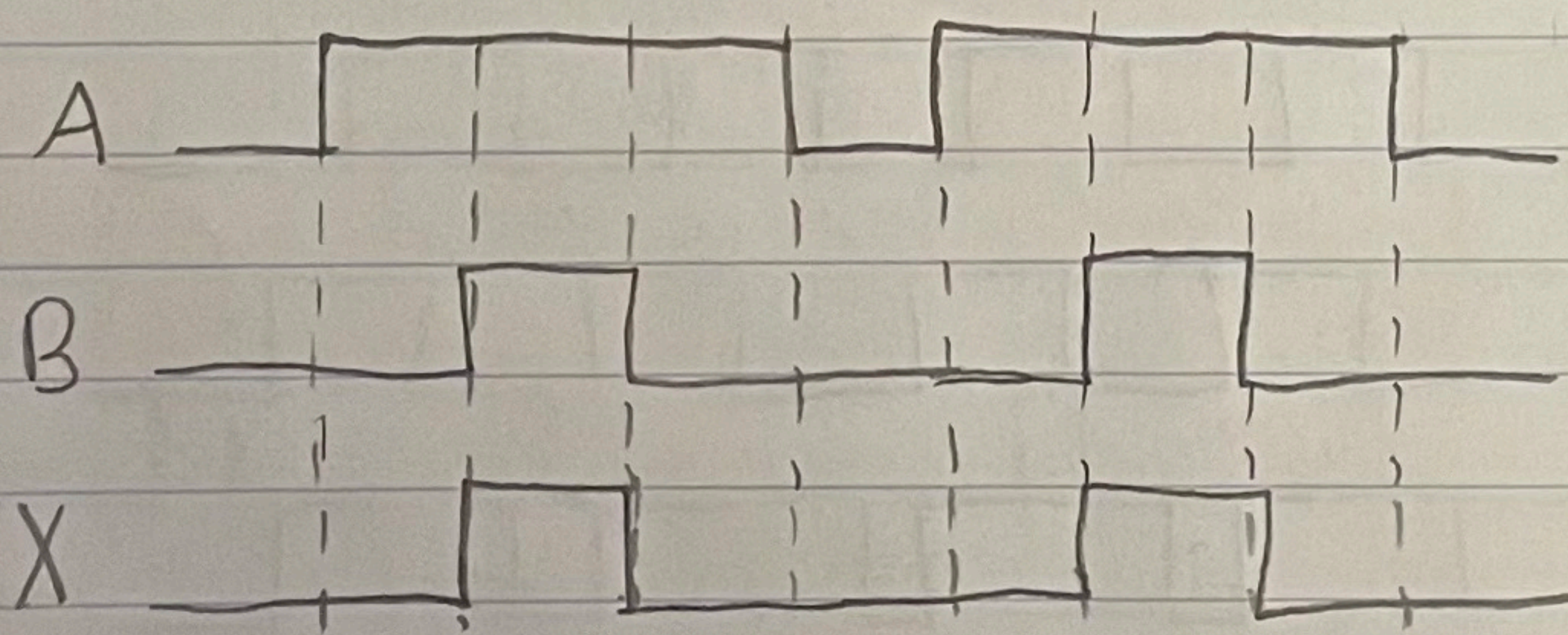
Homework 5

Ch 3: 2, 5, 6, 11, 17, 22, 26

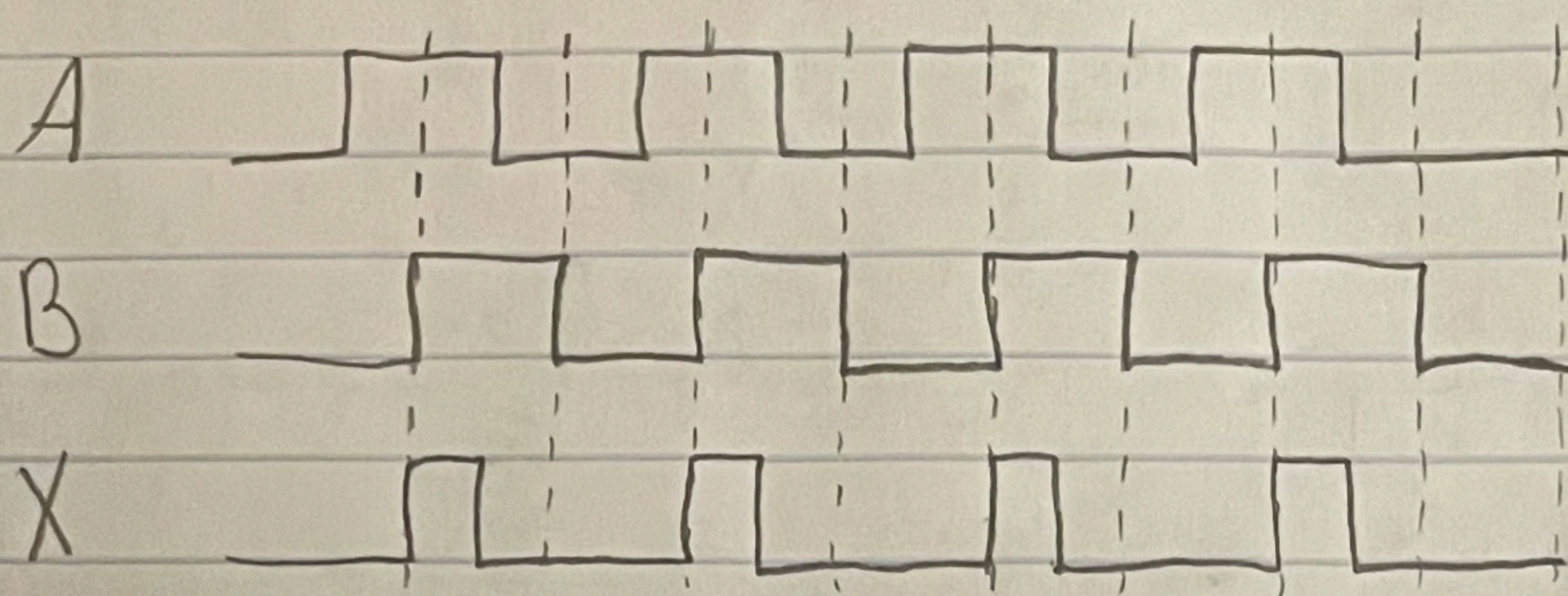
2 A combination of inverters is shown in Fig 3-77. If a HIGH is applied to point A, determine the logic levels at points B-F.



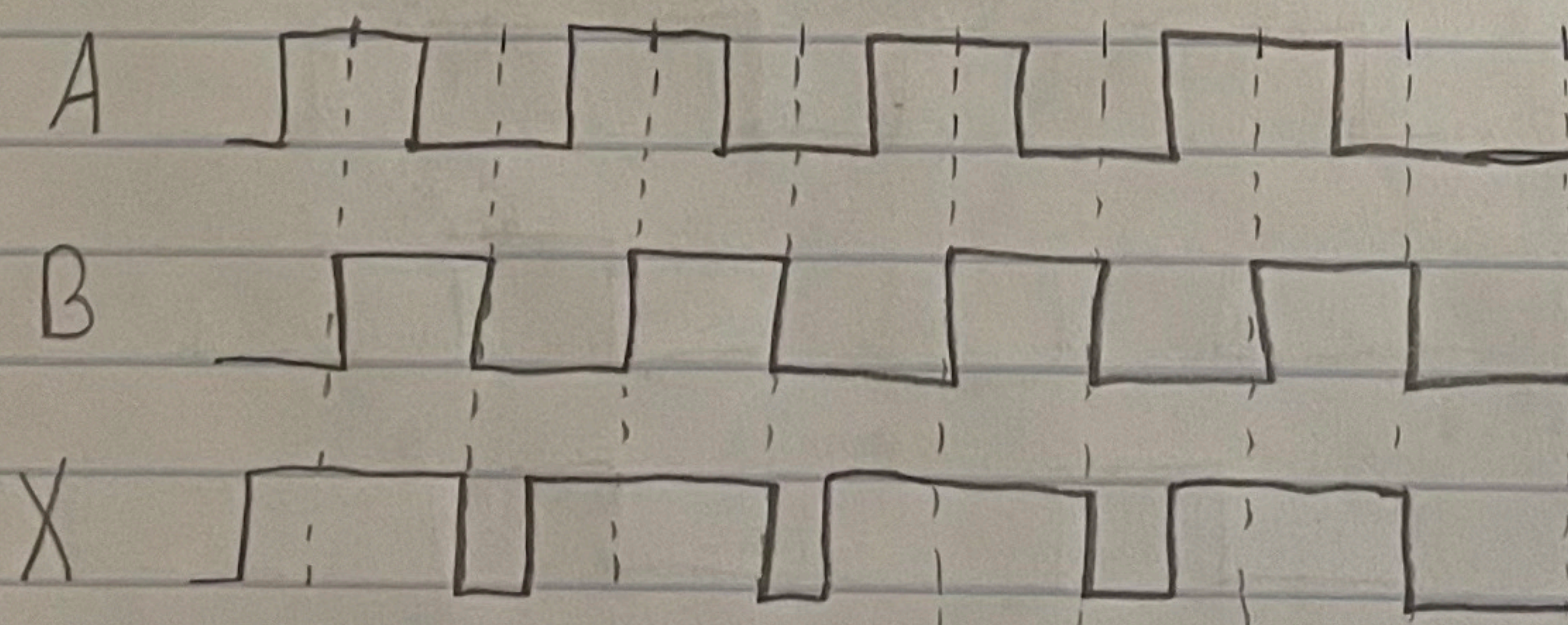
5 Determine the output, X, for a 2-input AND gate with the input waveforms shown in figure 3-7. Show the proper relationship of output to inputs with a timing diagram.



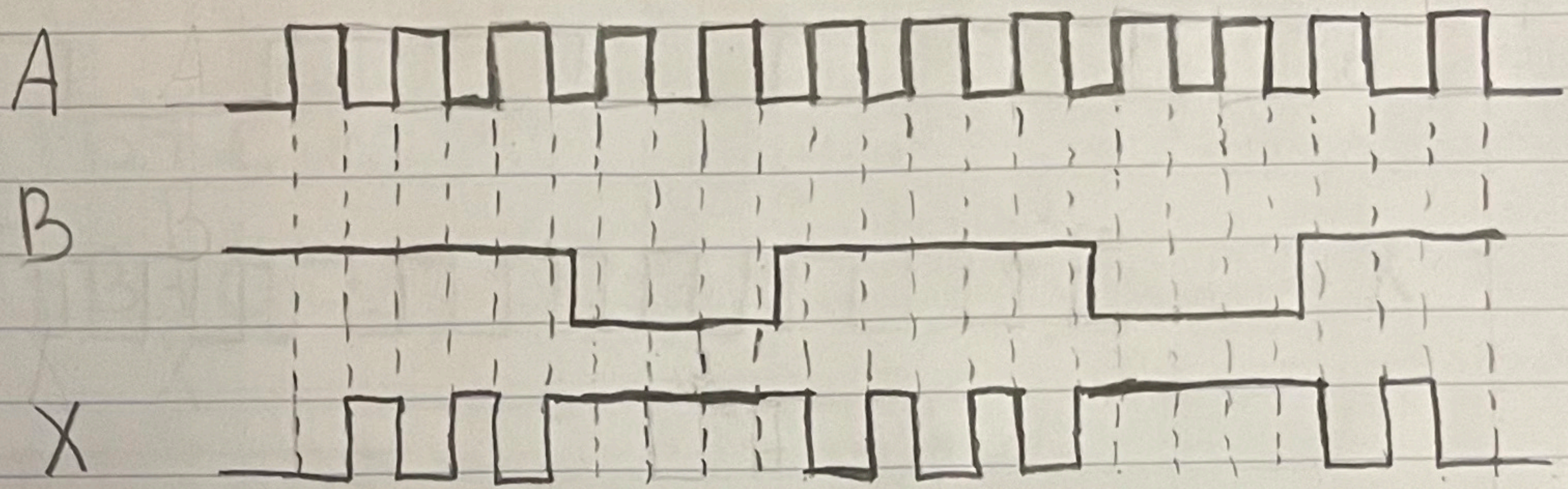
6 Repeat Problem 5 for the waveforms in Figure 3-79



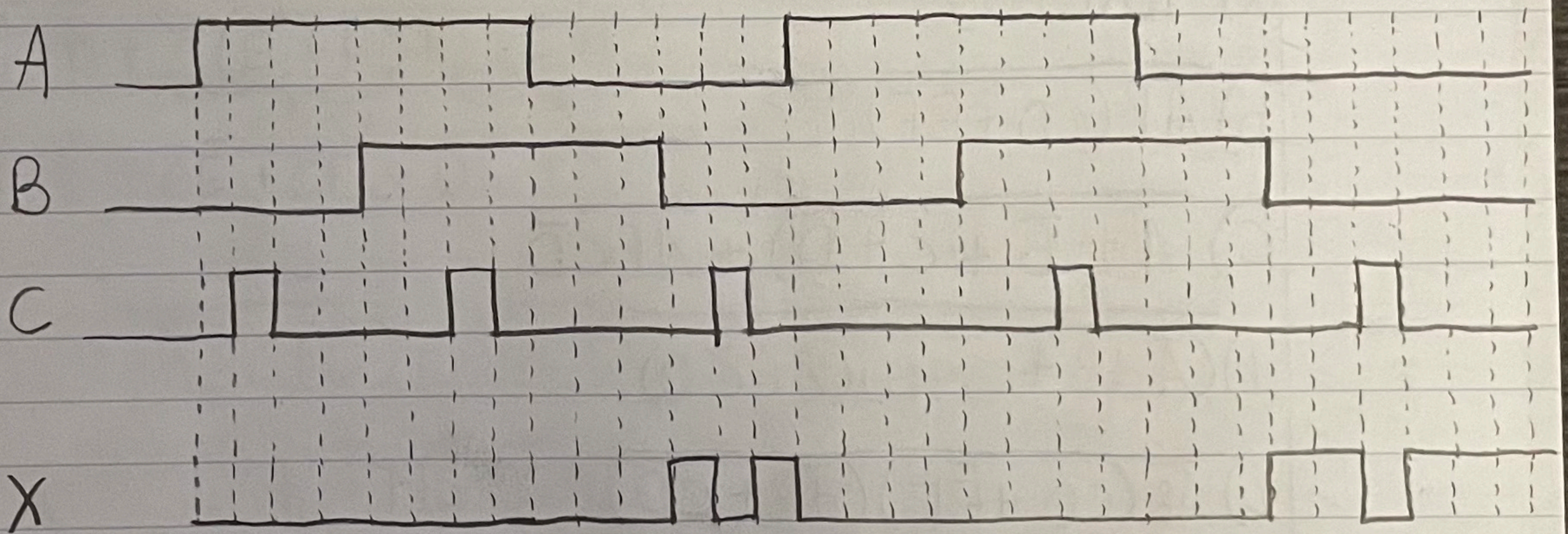
11 Determine the output for a 2 input OR gate when the input waveforms are as in Figure 3-79 and draw a timing diagram



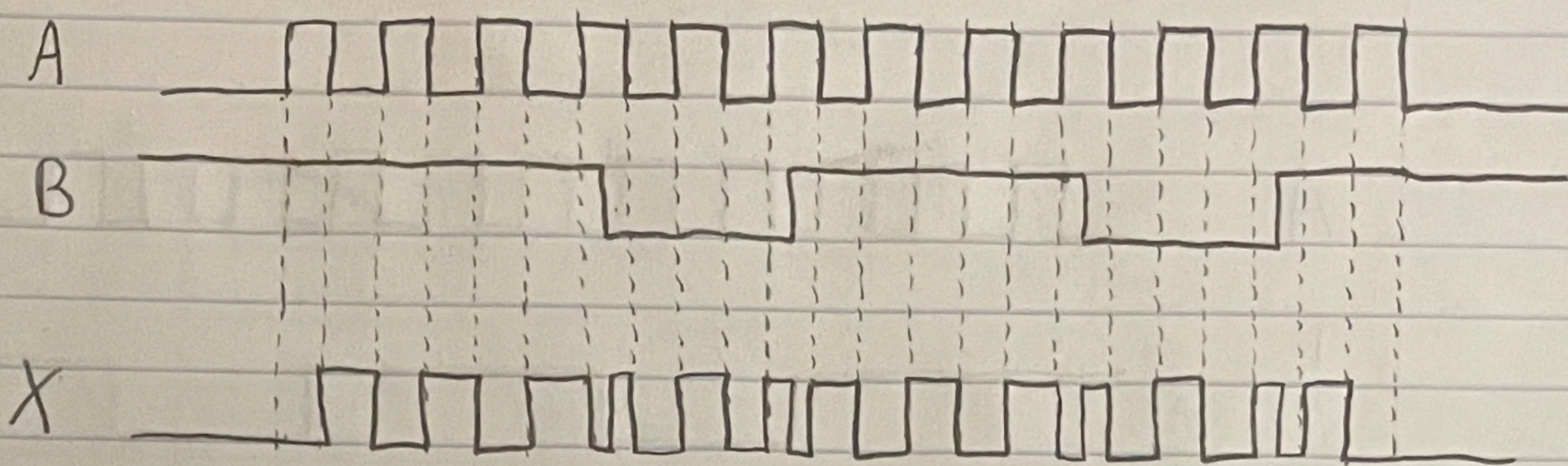
17 For the set of input waveforms in Figure 3-83, determine the output for the gate shown and draw the timing diagram



22 Determine the output waveform in Figure 3-87 and draw the timing diagram



26 Repeat Problem 17 for an exclusive OR gate



ch 4: 10, 15

to Apply DeMorgan's theorems to each expression:

a) $\overline{AB(c+\bar{D})}$

b) $\overline{AB(CD+EF)}$

c) $\overline{A + \bar{B} + C + \bar{D}} + \overline{ABCD}$

d) $\overline{(\bar{A} + B + C + D)(\bar{A}\bar{B}\bar{C}D)}$

e) $\overline{AB(CD + \bar{E}F)(\bar{A}B + C\bar{D})}$

$$a) \overline{AB(C+\bar{D})}$$

$$\overline{A+B+(C+\bar{D})}$$

$$\overline{A+B+C+\bar{D}}$$

$$\overline{A+B+C}\bar{D}$$

$$\overline{A+B+C}D$$

$$\overline{AB} = \overline{A+B} = \bar{A}\bar{B}$$

$$\overline{\bar{A}} = A$$

$$\overline{(A+B)} = \bar{A}\bar{B}$$

$$\overline{\bar{A}} = A$$

$$b) \overline{AB(CD+EF)}$$

$$\overline{A+B+(CD+EF)}$$

$$\overline{A+B+C+D+E+F}$$

$$\overline{A+B+C(E+F)+D(E+F)}$$

$$\overline{A+B+C}\bar{E}\bar{F} + \overline{A+B+C}E\bar{F} + \overline{A+B+C}\bar{D}E\bar{F} + \overline{A+B+C}D E\bar{F}$$

$$\overline{AB} = \overline{A+B} = \bar{A}\bar{B}$$

$$\overline{AB} = \overline{A+B} \text{ and } \overline{A+B} = \bar{A}\bar{B}$$

distribute

distribute

$$c) \overline{(A+B+C+\bar{D})+AB\bar{C}\bar{D}}$$

$$\overline{A+B+C+\bar{D}+AB\bar{C}\bar{D}}$$

$$\overline{A+B+C+\bar{D}+AB\bar{C}\bar{D}}$$

$$\overline{A+B+C+\bar{D}}$$

$$\overline{AB} = \overline{A+B} \text{ and } \overline{A+B} = \bar{A}\bar{B}$$

$$\overline{\bar{A}} = A$$

$$A+AB = A$$

$$d) \overline{(\overline{A+B+C+D})(\overline{A\overline{B}\overline{C}\overline{D}})}$$

$$\overline{(\overline{A+B+C+D}) + (\overline{A\overline{B}\overline{C}\overline{D}})}$$

$$(\overline{A+B+C+D}) + (\overline{A\overline{B}\overline{C}\overline{D}})$$

$$\overline{A+B+C+D}$$

$$\overline{AB} = \overline{A} + \overline{B}$$

$$\overline{\overline{A}} = A$$

$$A + AB = A$$

$$e) \overline{AB(CD+\overline{E}\overline{F})(\overline{A}\overline{B}+\overline{C}\overline{D})}$$

$$\overline{\overline{AB} + (\overline{CD+\overline{E}\overline{F}}) + (\overline{A\overline{B}+\overline{C}\overline{D}})}$$

$$AB + (\overline{C}+\overline{D})(\overline{E}+\overline{F}) + \overline{A\overline{B}\overline{C}\overline{D}}$$

$$AB + (\overline{C}+\overline{D})(\overline{E}+\overline{F}) + \overline{A\overline{B}\overline{C}\overline{D}}$$

$$AB + (\overline{C}+\overline{D})(\overline{E}+\overline{F})$$

$$AB + \overline{C}(\overline{E}+\overline{F}) + \overline{D}(\overline{E}+\overline{F})$$

$$AB + \overline{C}\overline{E} + \overline{C}\overline{F} + \overline{D}\overline{E} + \overline{D}\overline{F}$$

$$\overline{AB} = \overline{A} + \overline{B}$$

$$\overline{AB} = \overline{A} + \overline{B} \text{ and } A\overline{B} = \overline{\overline{A}\overline{B}}$$

$$\overline{\overline{A}} = A$$

$$A + AB = A$$

distribute

distribute

15 Draw the logic circuit represented by each expression:

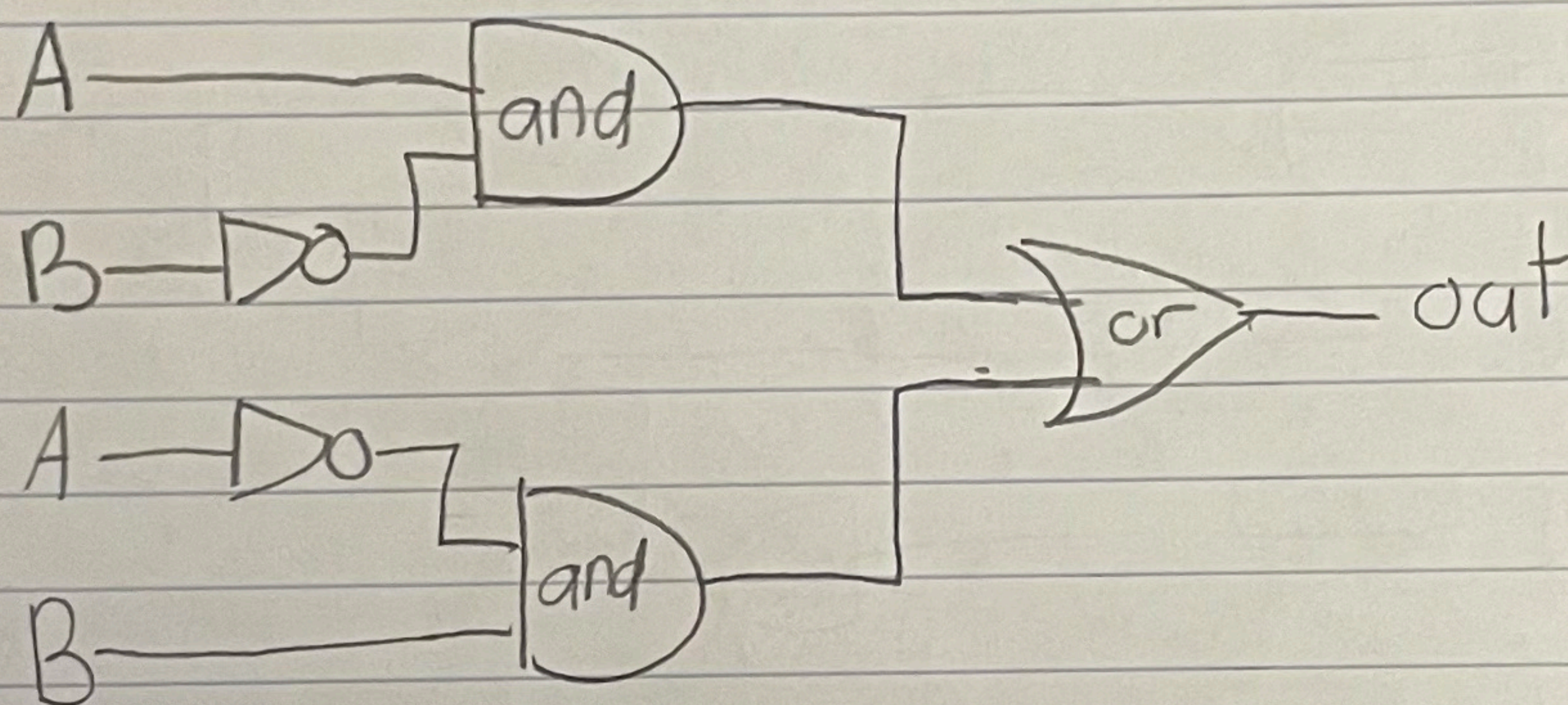
a) $A\bar{B} + \bar{A}B$

b) $AB + \bar{A}\bar{B} + \bar{A}BC$

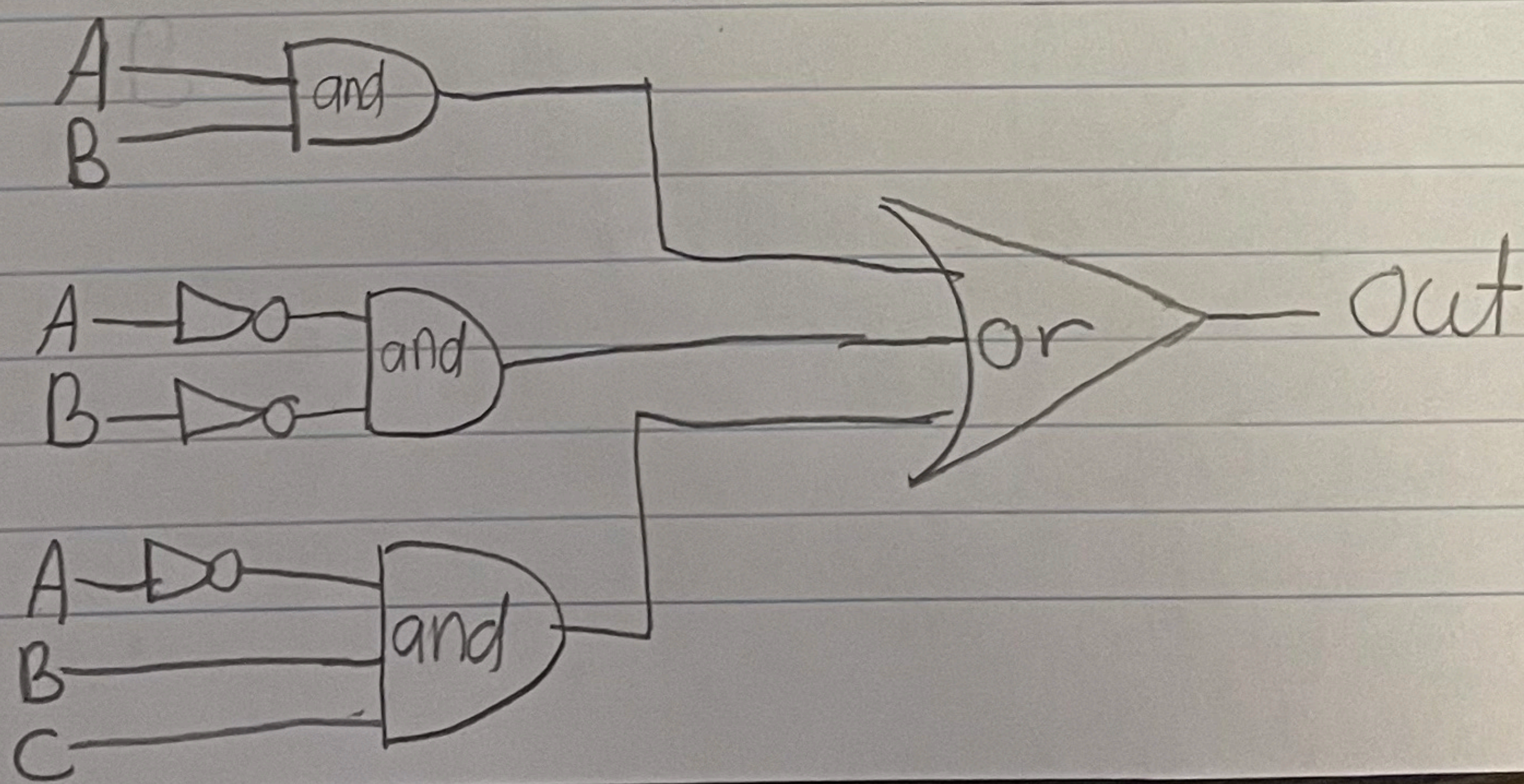
c) $\bar{A}B(C + \bar{D})$

d) $A + B(C + D(B + \bar{C}))$

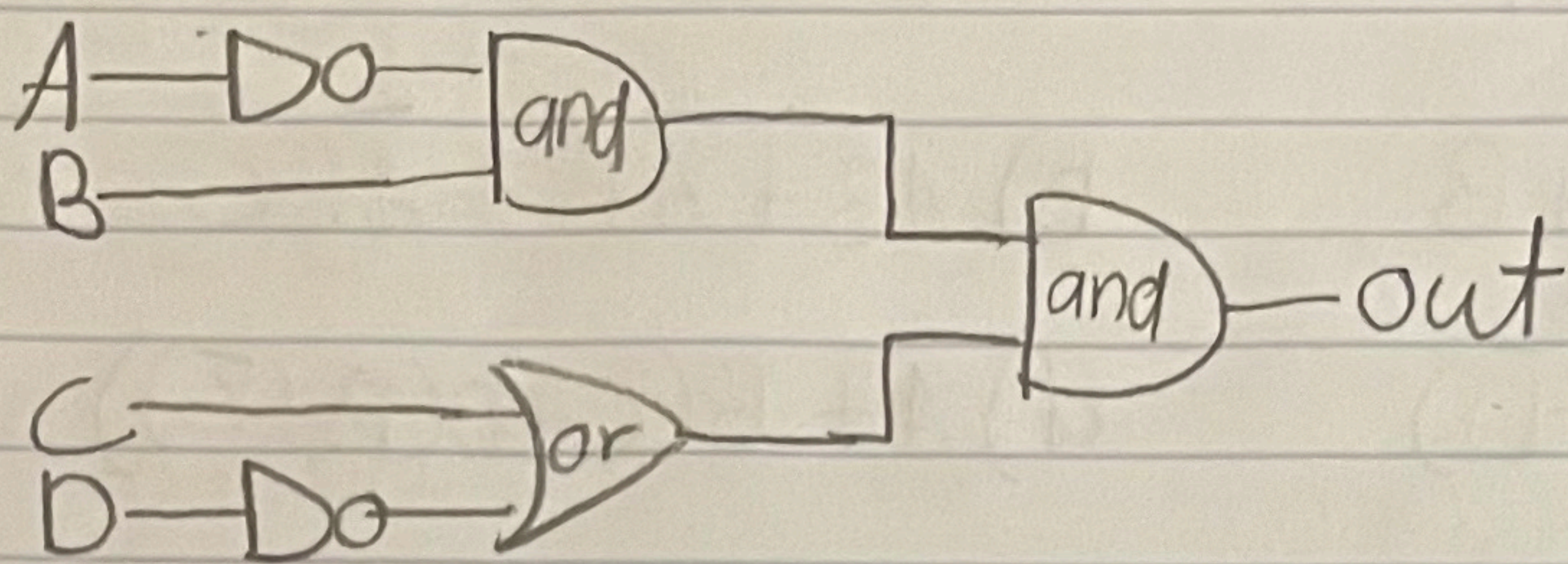
a) $A\bar{B} \text{ or } \bar{A}B$
and and



b) $AB + \bar{A}\bar{B} + \bar{A}BC$



c) $\bar{A} B (C + \bar{D})$



d) $A + B (C + D (B + \bar{C}))$

